Chemistry • Organic compounds and alkanes

Hydrocarbons

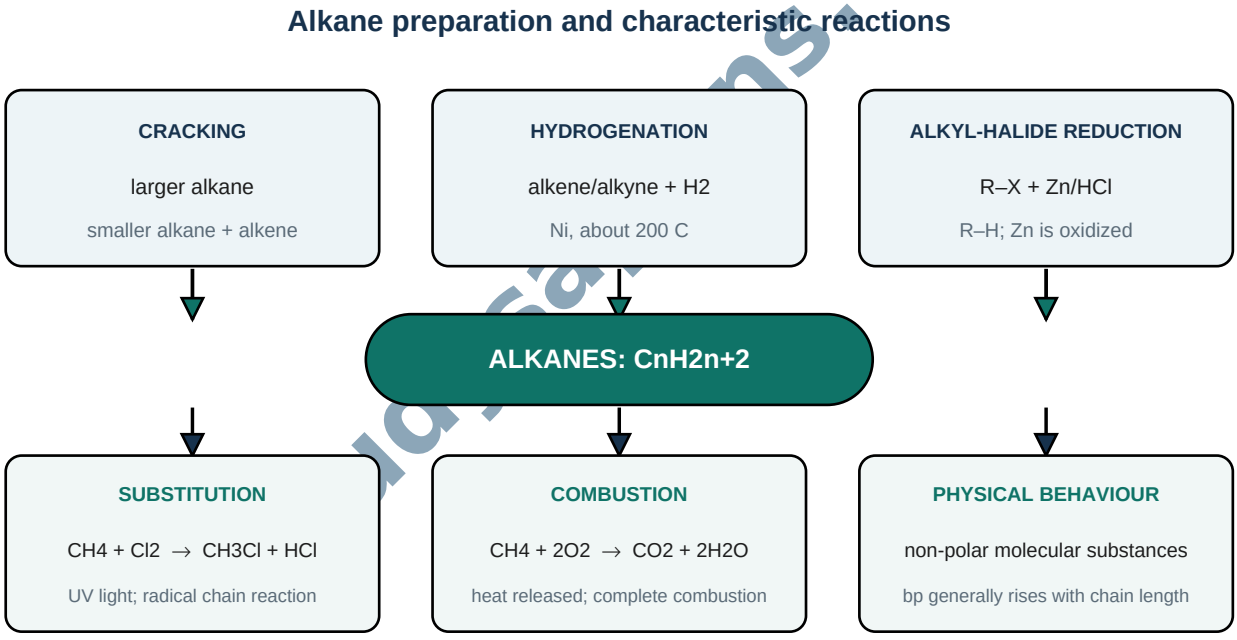
Official exercise • Complete solved answers • IUPAC and reaction map

Chapter 11 • Complete official exercise • 10 MCQs + 9 short + 6 constructed + 5 descriptive + 3 investigative

What this chapter is about

• **Organic chemistry** is the chemistry of carbon compounds; conventional inorganic exceptions include carbon oxides, carbonates, cyanides and carbides.
• **Alkanes** are saturated hydrocarbons with general formula C_nH_{2n+2} and only C—C and C—H single bonds.
• **Preparation:** cracking, catalytic hydrogenation of alkenes/alkynes and reduction of alkyl halides.
• **Reactions:** photochemical halogen substitution and combustion; alkanes do not normally undergo addition.

Figure 1 — Original reaction map for alkane preparation, substitution and combustion



Alkanes contain strong, nearly non-polar sigma bonds, so they do not undergo addition like alkenes.

Textbook clarification: Iso-butane is 2-methylpropane, C_4H_{10}. A five-carbon structure labelled “iso-butane” in the chapter discussion is actually 2-methylbutane (isopentane).

A. Multiple Choice Questions

1. Which other atom is almost always present along with carbon atom in all organic compounds?

(a) oxygen
(b) nitrogen
(c) hydrogen
(d) halogen

Answer: (c) hydrogen — Hydrogen is the characteristic partner of carbon in hydrocarbons and most other organic compounds.

2. Which other metal can be used to reduce alkyl halides?

(a) Al
(b) Mg
(c) Ni
(d) Co

Answer: (b) Mg — The intended answer is Mg: an active metal can be used with acid to supply hydrogen for replacement of halogen by H. Ni is the catalyst listed for alkene/alkyne hydrogenation.

3. If naphtha undergoes a combustion reaction, what products do you expect it to form?

(a) alkanes
(b) alkenes
(c) CO_2 and H_2O
(d) both alkanes and alkenes

Answer: (c) CO_2 and H_2O — Complete combustion of its hydrocarbon molecules produces carbon dioxide and water. Cracking, not combustion, produces smaller alkanes and alkenes.

4. Why does a mixture of zinc and hydrochloric acid act as a reducing agent?

(a) because zinc acts as a reducing agent
(b) because atomic hydrogen is produced with Zn/HCl which acts as a reducing agent
(c) because molecular hydrogen is produced with Zn/HCl which acts as a reducing agent
(d) because chloride ions are produced with Zn/HCl which act as a reducing agent

Answer: (b) because atomic hydrogen is produced with Zn/HCl which acts as a reducing agent — The book's intended answer is (b), using its older '[H]' model. Modern equations write $Zn + 2HCl \rightarrow ZnCl_2 + H_2$; zinc is oxidized and H^+ is reduced. The overall Zn/HCl system reduces R–X to R–H.

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5. Which alkane will evolve the most amount of heat when it is burnt with oxygen?

(a) ethane (b) propane
(c) n-butane (d) iso-butane

Answer: (c) n-butane — For equal moles, C_4 fuels release much more heat than ethane or propane. n-Butane is very slightly more exothermic than its more stable isomer iso-butane; both have formula C_4H_{10}, so the last pair is a close comparison.

6. Which reaction is not given by alkanes?

(a) substitution (b) combustion
(c) addition (d) cracking

Answer: (c) addition — Alkanes are saturated and have no C=C or C≡C π bond at which a normal addition reaction can occur.

7. Which hydrocarbon is responsible for explosions in coal mines?

(a) butane (b) pentane
(c) methane (d) ethene

Answer: (c) methane — Methane, historically called firedamp in mining, can form an explosive mixture with air and ignite.

8. Which product will be formed when ethyl bromide (CH_3CH_2Br) is treated with Zn/HCl?

(a) CH_4 (b) $CH_3—CH_3$
(c) $CH_3—CH_2—CH_2—CH_3$ (d) $CH_3—CH_2—CH_3$

Answer: (b) $CH_3—CH_3$ — Reduction replaces Br by H without changing the two-carbon skeleton: $CH_3CH_2Br \rightarrow CH_3CH_3$ (ethane).

9. Which of the following is not a process of halogenation of alkanes?

(a) cracking (b) chlorination
(c) bromination (d) iodination

Answer: (a) cracking — Cracking breaks larger hydrocarbons into smaller ones. Chlorination, bromination and iodination introduce halogens by substitution.

10. How many moles of oxygen will be required to completely burn propane?

(a) 4 moles (b) 5 moles
(c) 3 moles (d) 6 moles

Answer: (b) 5 moles — $C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O$, so one mole of propane requires five moles of O_2.

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B. Questions for Short Answers

11.1. Differentiate between an organic and an inorganic compound.

Feature	Typical organic compound	Typical inorganic compound
Essential basis	Contains carbon, usually bonded to H and often O, N or halogens	May contain any elements; carbon is not essential
Bonding/structure	Mainly covalent molecules or covalent networks; many isomers	Often ionic, metallic or simple covalent substances
Melting/boiling behaviour	Molecular examples often relatively low	Ionic/network examples often high
Water solubility	Hydrocarbons usually insoluble; polar examples may dissolve	Many salts dissolve, though many oxides/sulphides do not
Examples	CH_4, $C_6H_{12}O_6$, CH_3COOH	NaCl, H_2O, $CaCO_3$, CO_2

These are useful generalizations, not absolute rules. CO, CO_2, carbonates, cyanides and carbides are conventionally treated as inorganic despite containing carbon.

11.2. Why are organic compounds found in large numbers?

Carbon is tetravalent and forms strong covalent bonds with itself (**catenation**) and with H, O, N, halogens and other elements. It can make long or branched chains and rings, single/double/triple bonds and many isomers. Different functional groups and three-dimensional arrangements multiply the possibilities further.

11.3. Name the products which are obtained when natural gas is oxidized under controlled conditions.

Natural gas is mainly methane. With carefully controlled oxygen, temperature, pressure and catalyst, **partial oxidation** can produce useful chemicals instead of only CO_2 and H_2O:

$$CH_4 + \tfrac{1}{2}O_2 \rightarrow CH_3OH \text{ (methanol)}$$

$$CH_4 + O_2 \rightarrow HCHO + H_2O \text{ (methanal/formaldehyde)}$$

$$2CH_4 + O_2 \rightarrow 2CO + 4H_2 \text{ (synthesis gas)}$$

The exact industrial product depends on the process; severe or insufficiently controlled oxidation gives CO, carbon black/soot, CO_2 and water.

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11.4. How is naphtha fraction decomposed to give lower hydrocarbons?

Naphtha is **cracked** by strong heating, in the book's example at about 500°C over a zeolite catalyst. Carbon–carbon bonds break to produce shorter alkanes and alkenes with greater fuel or chemical-feedstock value.

Example: $C_{16}H_{34} \rightarrow C_8H_{18} + C_8H_{16}$

Naphtha (mixture) —heat / zeolite→ smaller alkanes + alkenes

11.5. Write down the molecular formula, structural formula and condensed structural formula for iso-butane.

Iso-butane = 2-methylpropane.

Molecular formula: C_4H_{10}

Structural/condensed connectivity: $CH_3—CH(CH_3)—CH_3$

Fully condensed structural formula: $(CH_3)_3CH$

The central CH carbon is bonded to three methyl groups and one hydrogen. This corrects the chapter's earlier five-carbon mislabel.

11.6. How are organic compounds useful for us?

They provide food (carbohydrates, proteins, lipids and vitamins), fuels, medicines, fibres and clothing, plastics, rubber, dyes, detergents, perfumes, solvents, pesticides and many biological materials such as DNA and enzymes. Their usefulness depends on molecular structure and responsible handling.

11.7. Write down the names of five such organic compounds which exist naturally.

Examples are **methane** (CH_4), **glucose** ($C_6H_{12}O_6$), **sucrose** ($C_{12}H_{22}O_{11}$), **ethanol** (C_2H_5OH, formed by fermentation), and **citric acid** ($C_6H_8O_7$). Natural occurrence does not automatically mean a substance is safe in every dose.

11.8. Give the IUPAC name to $CH_3—CH(CH_3)—CH_2—CH_3$.

1. The longest continuous chain has four carbons, so the parent is **butane**.

2. Number from the left to give the methyl substituent position 2 rather than 3.

3. The name is **2-methylbutane** (common name: isopentane).

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11.9. How do the melting and boiling points of alkanes change when we move from lower members to higher members?

They generally increase because molecular size, molar mass, electron-cloud surface area and London dispersion attraction increase. Boiling points show this trend especially clearly; melting points are less regular because crystal packing also matters. Among isomers, greater branching usually lowers boiling point by reducing contact area.

C. Constructed Response Questions

11.1. Why do alkanes show little reactivity towards other reagents?

Their C—C and C—H bonds are strong σ bonds and are nearly non-polar. Alkanes have no π bond, lone pair or strongly charged reaction centre for common acids, bases and oxidizing agents to attack under ordinary conditions. They nevertheless burn readily when ignited and substitute with halogens under UV light.

11.2. Why does a mixture of natural gas and air explode?

Leaked methane mixes with oxygen in air. Within its flammable range (approximately 5–15% methane by volume in air), a spark, flame or hot surface starts extremely rapid combustion.

$$CH_4(g) + 2O_2(g) \rightarrow CO_2(g) + 2H_2O(g) + \text{heat}$$

Heat makes gases expand rapidly; in an enclosed kitchen or mine the fast pressure rise can be explosive. Too little methane cannot sustain combustion, while too little oxygen also prevents it.

11.3. How do you compare the melting and boiling points of inorganic and organic compounds?

Molecular organic compounds often have lower melting and boiling points because weak intermolecular forces must be overcome, whereas many ionic inorganic compounds have strong electrostatic lattices and high values. This is only a trend: diamond and cross-linked organic networks have very high melting behaviour, while inorganic molecular substances such as CO_2 and H_2O have relatively low values.

11.4. Reactions of alkanes with chlorine take place in the presence of sunlight. What is the role of sunlight?

UV energy causes **homolytic fission** of Cl—Cl and starts a free-radical chain reaction.

$$\text{Initiation: } Cl_2 \xrightarrow{UV} 2Cl\bullet$$

$$\text{Propagation: } Cl\bullet + CH_4 \rightarrow HCl + CH_3\bullet$$

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$$CH_3\bullet + Cl_2 \rightarrow CH_3Cl + Cl\bullet$$

The regenerated chlorine radical continues the chain. Further substitution can form CH_2Cl_2, $CHCl_3$ and CCl_4.

11.5. How do you compare the boiling point of n-butane with that of iso-butane?

n-Butane has the higher boiling point (about −0.5°C versus about −11.7°C for iso-butane/2-methylpropane). The straight chain has a larger effective contact surface and stronger London dispersion forces; the compact branched molecule has weaker intermolecular attraction.

11.6. Why are organic compounds not generally soluble in water?

Many organic compounds, especially hydrocarbons, are non-polar and cannot form strong hydrogen bonds or ion–dipole attractions with water. Water therefore prefers interactions with itself ("like dissolves like"). Organic molecules with sufficiently strong polar/ionic groups—such as small alcohols, sugars and carboxylate salts—can be water-soluble.

D. Descriptive Questions

11.1. Describe the importance of organic compounds in daily life.

Area	Examples and importance
Nutrition and life	Glucose, starch, proteins, fats, vitamins, DNA and enzymes store energy or build/regulate living systems.
Fuels	Methane, LPG, petrol, diesel and kerosene supply energy for cooking, transport and industry.
Medicine and hygiene	Paracetamol, antibiotics, antiseptics, soaps and detergents prevent or treat disease and improve sanitation.
Clothing and materials	Cotton, wool, silk, polyester, nylon, plastics, rubber and paints provide fabrics and manufactured goods.
Agriculture	Fertilizer-related organics, pesticides and plant-growth chemicals can support crop production when safely used.
Household/industry	Vinegar, fragrances, dyes, adhesives, solvents and cosmetics perform many specialized functions.

Benefits must be balanced against toxicity, pollution, fire risk and waste. Safe dosage, ventilation, labelled storage and proper disposal are essential.

11.2. Why is carbon so important as an element that a whole branch of chemistry is based on it?

• **Tetravalency:** Four covalent bonds give many stable bonding arrangements.

• **Catenation:** Strong C—C bonds allow long chains, branched chains and rings of many sizes.

• **Multiple bonding:** Carbon forms C=C and C≡C as well as single bonds, creating unsaturation and distinctive reactions.

• **bonding versatility:** It bonds strongly with H, O, N, S, halogens and many other elements.

• **Isomerism:** The same molecular formula can produce different connectivities, functional-group positions and spatial arrangements.

• **Biological compatibility:** Stable yet transformable carbon frameworks support proteins, carbohydrates, lipids and nucleic acids.

No other element combines these features to the same extent, which explains the enormous number and diversity of organic compounds.

11.3. A C—C single bond in alkanes does not behave as a functional group, but C=C in alkenes does. Explain.

A **functional group** is an atom, group or bond that gives a family of compounds characteristic chemical reactions. C=C contains a reactive π bond above and below the σ-bond framework. Its exposed electron density attracts electrophiles and allows addition of H_2, halogens, hydrogen halides and water under suitable conditions.

An ordinary alkane C—C single bond has only a strong, nearly non-polar σ bond and no comparable π-electron centre. Its usual reactions—combustion and radical halogen substitution—do not make each C—C bond a characteristic functional-group site. Thus C=C defines the alkene family; C—C alone is part of the ordinary hydrocarbon skeleton.

11.4. Explain the IUPAC system of nomenclature for alkanes.

An alkane name combines a **prefix** (substituent and position), a **root** (number of carbons in the parent chain) and the suffix **—ane**.

Carbons	Root	Carbons	Root
1	meth-	6	hex-
2	eth-	7	hept-
3	prop-	8	oct-
4	but-	9	non-

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Carbons	Root	Carbons	Root
5	pent-	10	dec-

1. Choose the **longest continuous carbon chain** as the parent.

2. Identify alkyl branches: CH_3— is methyl, C_2H_5— is ethyl, etc.

3. Number from the end giving the substituent the **lowest locant** (or lowest set of locants for several branches).

4. Write position–branch + parent: use hyphens between numbers and words and commas between numbers.

5. For repeated identical branches use di-, tri-, tetra-; list different substituents alphabetically.

Example: CH_3—$CH(CH_3)$—CH_2—CH_3 = 2-methylbutane. CH_3—$CH(CH_3)$—CH_3 = 2-methylpropane (iso-butane).

11.5. How is combustion reaction of alkanes useful for us?

Alkane combustion is exothermic. It supplies heat for cooking and industry, mechanical energy in engines and electricity through thermal power generation.

$$\text{General: } C_nH_{2n+2} + (3n+1)/2\ O_2 \rightarrow nCO_2 + (n+1)H_2O + \text{heat}$$

$$CH_4 + 2O_2 \rightarrow CO_2 + 2H_2O + \text{heat}$$

$$C_3H_8 + 5O_2 \rightarrow 3CO_2 + 4H_2O + \text{heat}$$

Bond breaking in fuel/O_2 requires energy, but formation of strong C=O and O—H bonds releases more, giving a net energy output.

Safe use requires enough oxygen and ventilation. Incomplete combustion can produce toxic CO and soot; all combustion of fossil hydrocarbons adds CO_2, and leaks can form explosive mixtures.

E. Investigative Questions

11.1. A natural-gas valve is left open and gas spreads through the kitchen. Why may it explode, and how can this be avoided?

Methane diffuses and mixes with air. Once the mixture is in its flammable range, even a small ignition source can trigger rapid combustion and a damaging pressure wave.

• **Prevention:** Close valves, maintain pipes/regulators, ventilate kitchens, install a suitable gas detector and never ignore the odorant added to gas.

• **If a leak is suspected:** Do not use flames or operate electrical switches/appliances. If safe, close the supply and open ventilation without creating sparks.

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• **Evacuate** and call the gas emergency service or fire service from a safe location. Do not re-enter until declared safe.

• **Never test for a leak with a flame;** use approved leak-detection procedures by trained personnel.

11.2. “Neem” is commonly grown in Pakistan. Comment on its medicinal benefits.

Neem (*Azadirachta indica*) contains organic compounds including **azadirachtin, nimbin and nimbidin**. Traditional practices use neem twigs for oral cleaning and preparations of leaves or bark for skin and other ailments. Laboratory studies report antibacterial, antifungal, anti-inflammatory and insect-antifeedant activity for some extracts.

These findings do not prove that every home preparation cures disease. Effectiveness and safety depend on species, plant part, extraction, dose and formulation. **Neem oil can be toxic if swallowed, especially to children, and unsupervised use in pregnancy or with medicines is unsafe.** Use regulated products and qualified medical advice rather than replacing prescribed treatment.

11.3. Name a few popular medicines which are, in fact, organic compounds.

Active ingredient	Molecular formula	Common therapeutic class/use
Paracetamol (acetaminophen)	$C_8H_9NO_2$	Analgesic and antipyretic
Aspirin (acetylsalicylic acid)	$C_9H_8O_4$	Analgesic/antipyretic; antiplatelet at prescribed doses
Ibuprofen	$C_{13}H_{18}O_2$	Non-steroidal anti-inflammatory drug
Amoxicillin	$C_{16}H_{19}N_3O_5S$	Beta-lactam antibiotic for susceptible bacterial infections
Salbutamol	$C_{13}H_{21}NO_3$	Bronchodilator used in asthma medicines

These are active ingredients; tablets or syrups also contain excipients. Organic origin does not guarantee safety—use only the correct medicine and dose under appropriate guidance.

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Points to Remember

Alkanes: C_nH_{2n+2}; saturated hydrocarbons with only C—C and C—H single bonds.

Iso-butane = 2-methylpropane = $(CH_3)_3CH$; 2-methylbutane has five carbons.

Cracking makes smaller alkanes and alkenes; hydrogenation adds H_2; Zn/HCl reduction replaces alkyl-halide halogen with H.

Chlorination is UV-initiated radical substitution and may continue to CH_2Cl_2, $CHCl_3$ and CCl_4.

Complete combustion gives CO_2 + H_2O + heat; limited oxygen can produce CO and soot.

Longer chains generally raise boiling point; branching lowers boiling point among isomers.

C=C is a functional group because its π bond gives characteristic addition reactions; ordinary C—C is not.

Methane–air mixtures can explode only within a flammable composition range and with an ignition source.

MCQ key at a glance

Q	Key	Q	Key	Q	Key	Q	Key	Q	Key
1	c	2	b	3	c	4	b	5	c
6	c	7	c	8	b	9	a	10	b

How to attempt this chapter in the paper

- Balance combustion before calculating O_2: propane needs 5 mol O_2 per mole of fuel.
- For IUPAC names, first find the longest chain, then number from the end nearest a substituent.
- Do not call cracking a combustion or halogenation reaction; compare both products and conditions.
- Distinguish the book's intended "[H]" notation from the modern oxidation explanation: Zn is oxidized.